

TP 7 : Apache hbase

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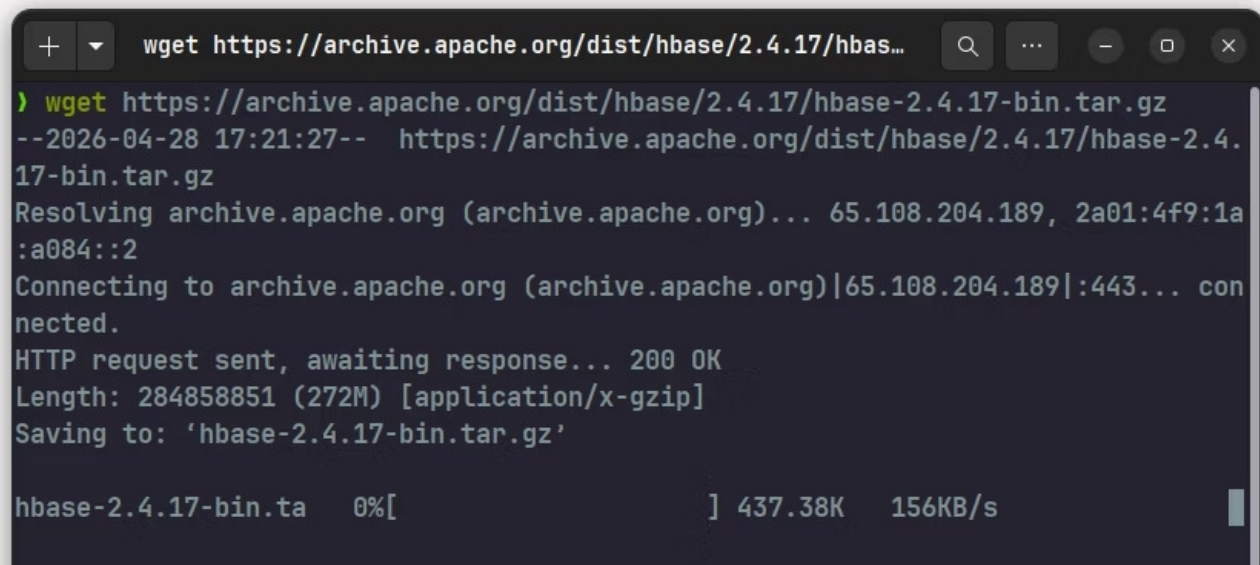
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Installation du Hbase

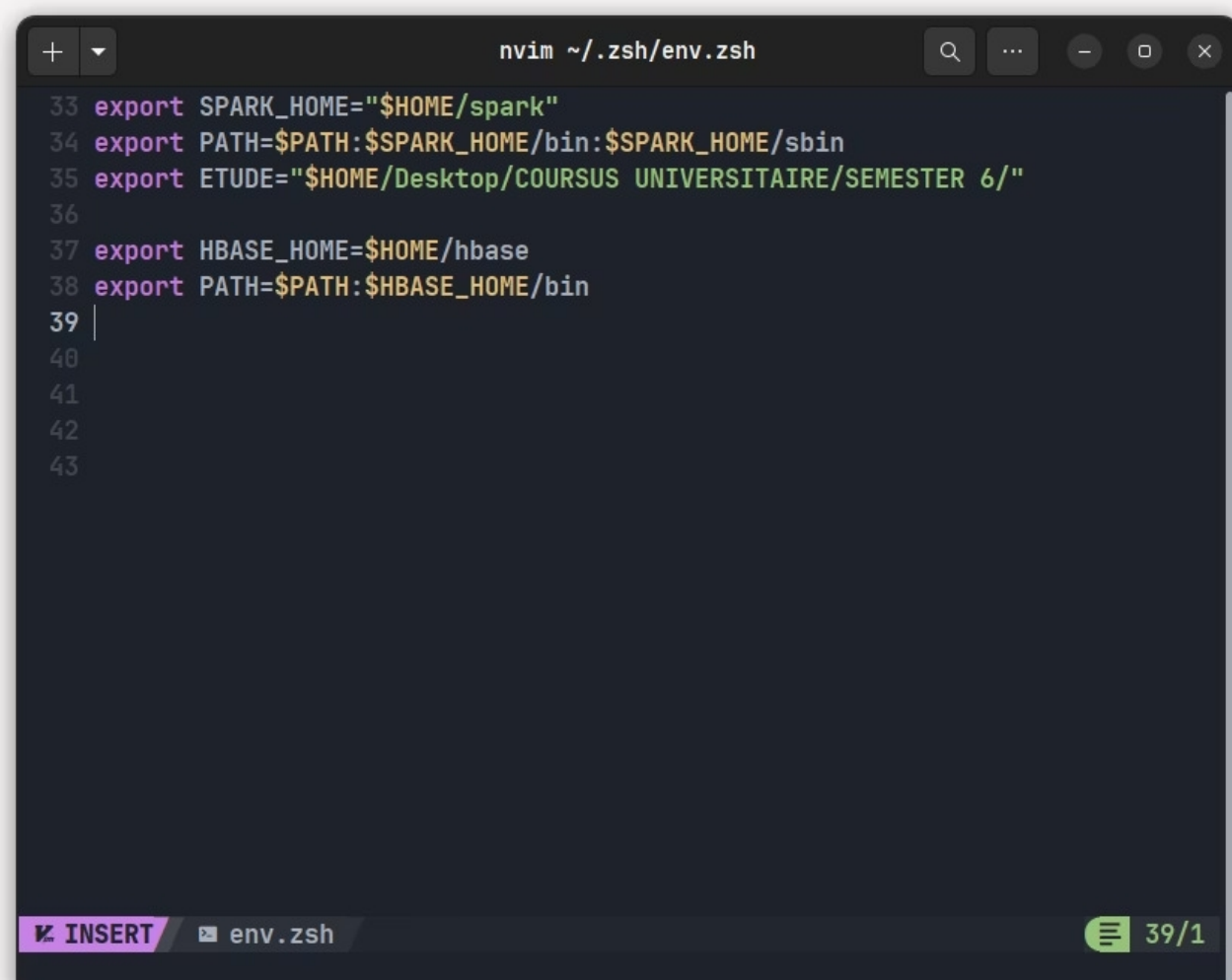
1. Téléchargement et extraction de fichier



```
wget https://archive.apache.org/dist/hbase/2.4.17/hbase-2.4.17-bin.tar.gz
--2026-04-28 17:21:27-- https://archive.apache.org/dist/hbase/2.4.17/hbase-2.4.17-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 284858851 (272M) [application/x-gzip]
Saving to: 'hbase-2.4.17-bin.tar.gz'

hbase-2.4.17-bin.ta  0%[                  ] 437.38K  156KB/s
```

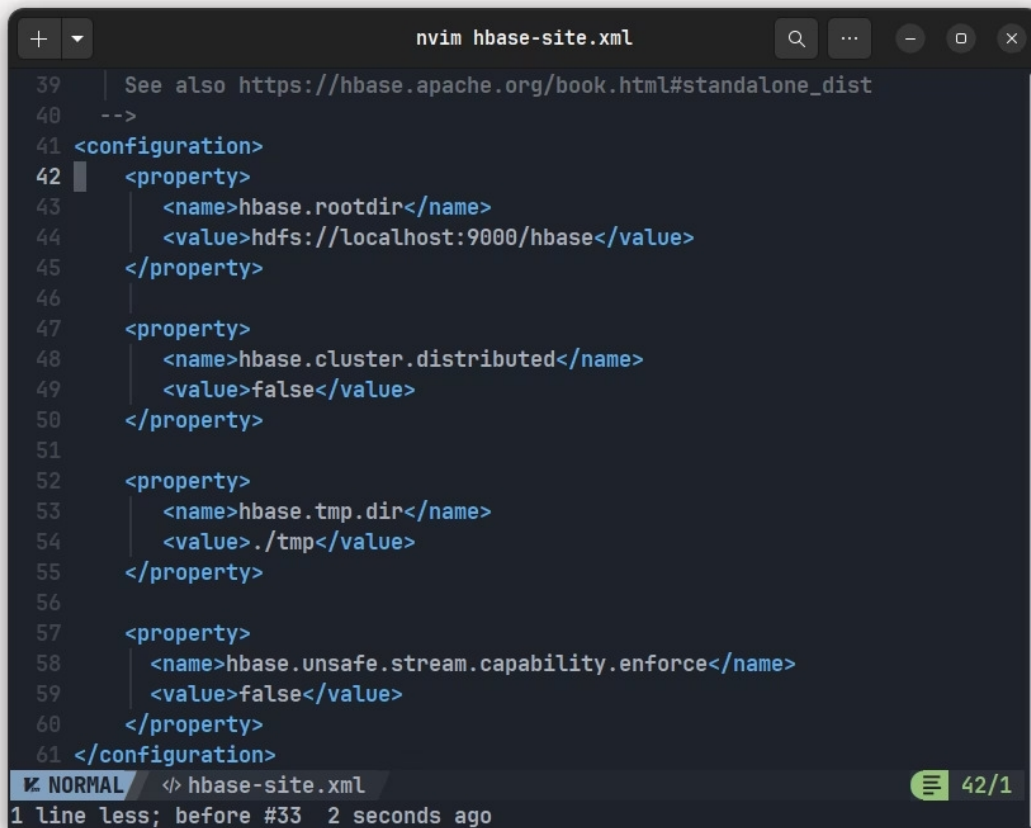
2. Configuration des variable d'environnement



```
nvim ~/.zsh/env.zsh
33 export SPARK_HOME="$HOME/spark"
34 export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
35 export ETUDE="$HOME/Desktop/COURSUS UNIVERSITAIRE/SEMESTER 6/"
36
37 export HBASE_HOME=$HOME/hbase
38 export PATH=$PATH:$HBASE_HOME/bin
39 |
40
41
42
43
```

INSERT env.zsh 39/1

3. Configuration du Hbase

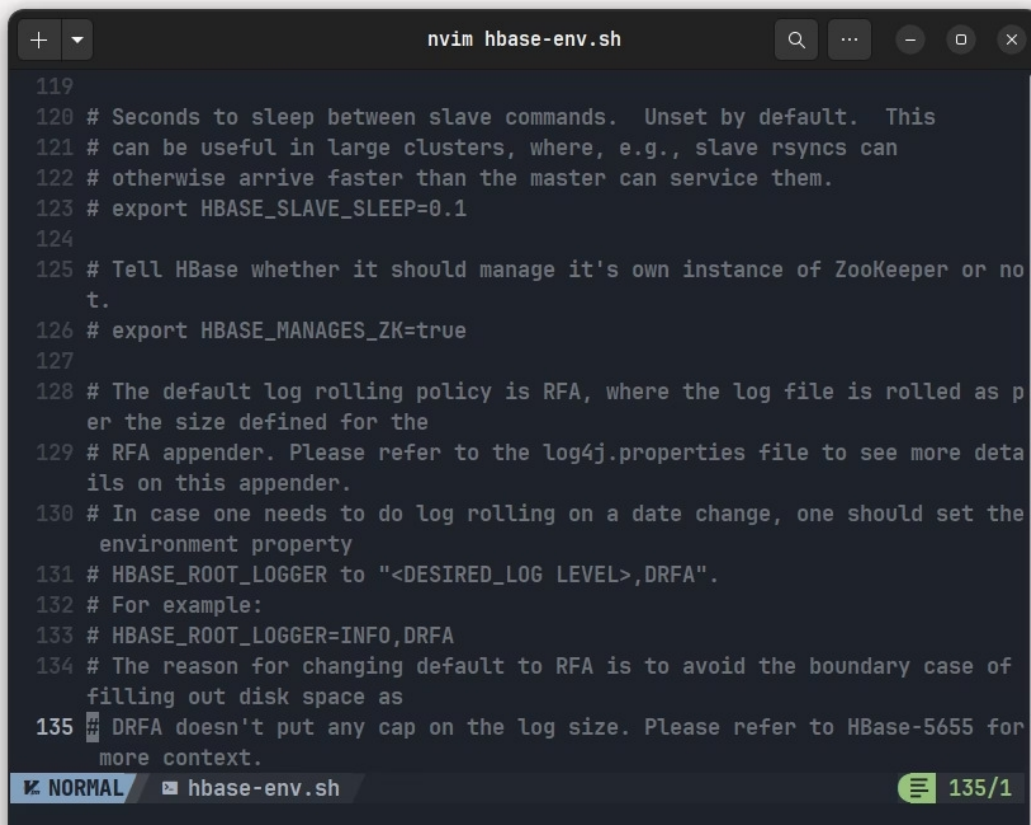


The screenshot shows the nvim editor with the file hbase-site.xml open. The editor is in NORMAL mode. The content of the file is an XML configuration for HBase. The configuration includes a <configuration> root element containing several <property> elements. The properties are: hbase.rootdir (value: hdfs://localhost:9000/hbase), hbase.cluster.distributed (value: false), hbase.tmp.dir (value: ./tmp), and hbase.unsafe.stream.capability.enforce (value: false). The status bar at the bottom shows 'NORMAL' mode, the file name 'hbase-site.xml', and a cursor position of '42/1'. A message '1 line less; before #33 2 seconds ago' is visible in the bottom left corner.

```
39 | See also https://hbase.apache.org/book.html#standalone_dist
40 -->
41 <configuration>
42 |   <property>
43 |     <name>hbase.rootdir</name>
44 |     <value>hdfs://localhost:9000/hbase</value>
45 |   </property>
46 |
47 |   <property>
48 |     <name>hbase.cluster.distributed</name>
49 |     <value>false</value>
50 |   </property>
51 |
52 |   <property>
53 |     <name>hbase.tmp.dir</name>
54 |     <value>./tmp</value>
55 |   </property>
56 |
57 |   <property>
58 |     <name>hbase.unsafe.stream.capability.enforce</name>
59 |     <value>false</value>
60 |   </property>
61 </configuration>
```

NORMAL </> hbase-site.xml 42/1
1 line less; before #33 2 seconds ago

4. installation du java



The screenshot shows the nvim editor with the file hbase-env.sh open. The editor is in NORMAL mode. The content of the file is a shell script for configuring HBase. It includes comments and export statements for HBASE_SLAVE_SLEEP, HBASE_MANAGES_ZK, and HBASE_ROOT_LOGGER. The status bar at the bottom shows 'NORMAL' mode, the file name 'hbase-env.sh', and a cursor position of '135/1'.

```
119
120 # Seconds to sleep between slave commands. Unset by default. This
121 # can be useful in large clusters, where, e.g., slave rsyncs can
122 # otherwise arrive faster than the master can service them.
123 # export HBASE_SLAVE_SLEEP=0.1
124
125 # Tell HBase whether it should manage it's own instance of ZooKeeper or no
126 # t.
127 # export HBASE_MANAGES_ZK=true
128
129 # The default log rolling policy is RFA, where the log file is rolled as p
130 # er the size defined for the
131 # RFA appender. Please refer to the log4j.properties file to see more deta
132 # ils on this appender.
133 # In case one needs to do log rolling on a date change, one should set the
134 # environment property
135 # HBASE_ROOT_LOGGER to "<DESIRED_LOG LEVEL>,DRFA".
136 # For example:
137 # HBASE_ROOT_LOGGER=INFO,DRFA
138 # The reason for changing default to RFA is to avoid the boundary case of
139 # filling out disk space as
140 # DRFA doesn't put any cap on the log size. Please refer to HBase-5655 for
141 # more context.
```

NORMAL hbase-env.sh 135/1

5. Démarage de Hadoop et Hbase

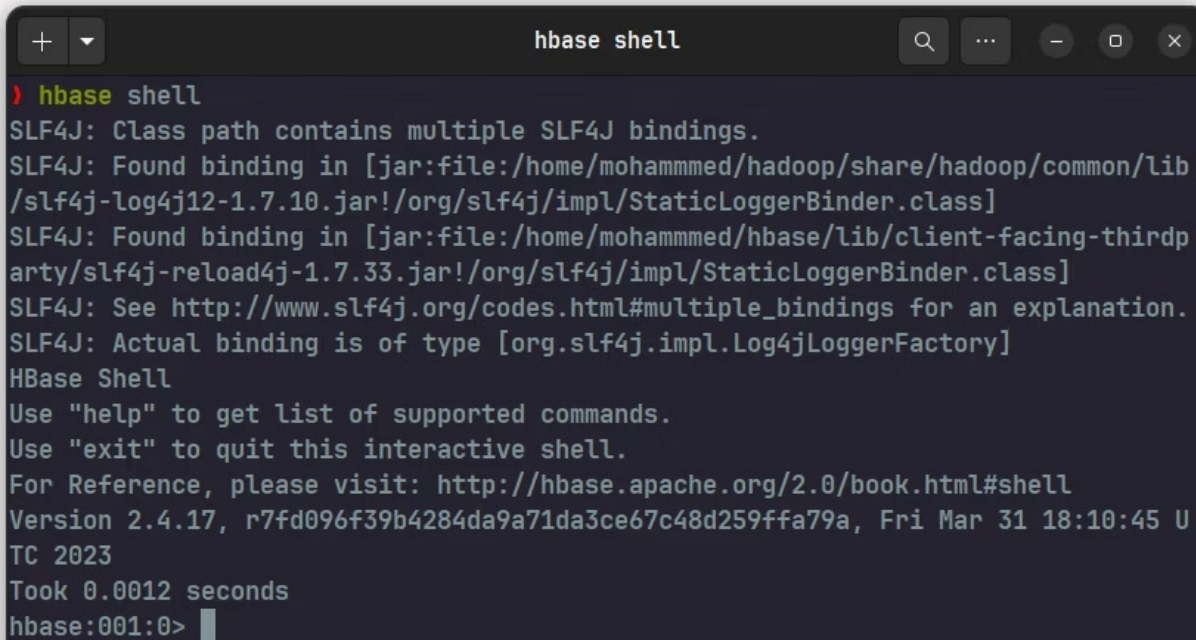
```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...  
> start-dfs.sh & start-yarn.sh & start-hbase.sh  
[1] 119222  
[2] 119223  
starting yarn daemons  
starting resourcemanager, logging to /home/mohammed/hadoop/logs/yarn-mohammed-  
resourcemanager-ROG-Strix.out  
SLF4J: Class path contains multiple SLF4J bindings.  
SLF4J: Found binding in [jar:file:/home/mohammed/hadoop/share/hadoop/common/lib  
/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: Found binding in [jar:file:/home/mohammed/hbase/lib/client-facing-thirdp  
arty/slf4j-reload4j-1.7.33.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.  
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]  
Starting namenodes on [localhost]  
running master, logging to /home/mohammed/hbase/logs/hbase-mohammed-master-ROG  
-Strix.out  
localhost: starting namenode, logging to /home/mohammed/hadoop/logs/hadoop-moha  
mmmed-namenode-ROG-Strix.out  
localhost: starting nodemanager, logging to /home/mohammed/hadoop/logs/yarn-moh  
ammed-nodemanager-ROG-Strix.out  
SLF4J: Class path contains multiple SLF4J bindings.  
SLF4J: Found binding in [jar:file:/home/mohammed/hadoop/share/hadoop/common/lib  
/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: Found binding in [jar:file:/home/mohammed/hbase/lib/client-facing-thirdp  
arty/slf4j-reload4j-1.7.33.jar!/org/slf4j/impl/StaticLoggerBinder.class]
```

6. Vérification

```
mohammed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE...  
> jps  
121169 Jps  
119842 NameNode  
120117 NodeManager  
119333 ResourceManager  
119565 HMaster  
120684 DataNode  
120911 SecondaryNameNode  
TP HBASE >
```

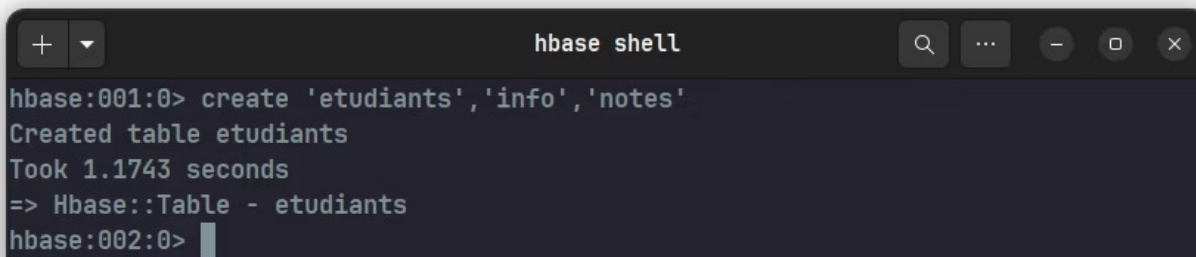

Manipulation des données

1. Lacement du shell



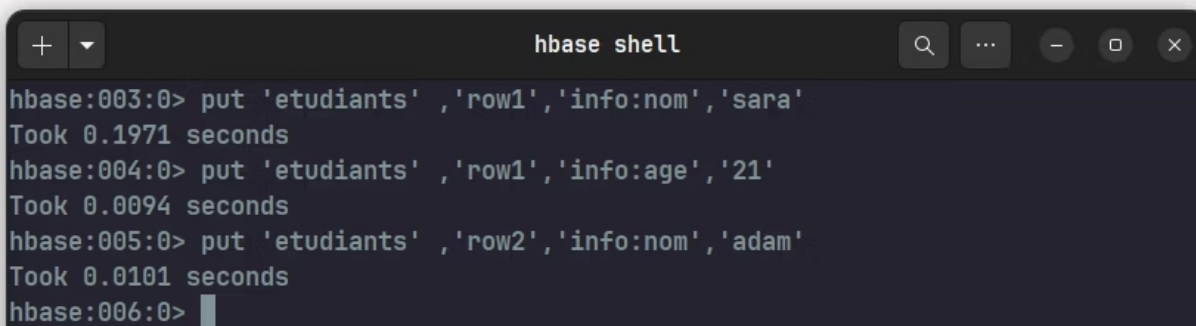
```
hbase shell
) hbase shell
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/mohammed/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/mohammed/hbase/lib/client-facing-thirdparty/slf4j-reload4j-1.7.33.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.4.17, r7fd096f39b4284da9a71da3ce67c48d259ffa79a, Fri Mar 31 18:10:45 UTC 2023
Took 0.0012 seconds
hbase:001:0>
```

2. Creation de la table



```
hbase:001:0> create 'etudiants','info','notes'
Created table etudiants
Took 1.1743 seconds
=> Hbase::Table - etudiants
hbase:002:0>
```

3. Insertion des données



```
hbase:003:0> put 'etudiants' , 'row1' , 'info:nom' , 'sara'
Took 0.1971 seconds
hbase:004:0> put 'etudiants' , 'row1' , 'info:age' , '21'
Took 0.0094 seconds
hbase:005:0> put 'etudiants' , 'row2' , 'info:nom' , 'adam'
Took 0.0101 seconds
hbase:006:0>
```

4. Lecture des données

```
hbase shell
hbase:006:0> get 'etudiants','row1'
COLUMN           CELL
info:age          timestamp=2026-04-28T19:07:42.066, value=21
info:nom          timestamp=2026-04-28T19:07:17.085, value=sara
1 row(s)
Took 0.1218 seconds
hbase:007:0> scan 'etudiants'
ROW              COLUMN+CELL
row1             column=info:age, timestamp=2026-04-28T19:07:42.066, value=21
row1             column=info:nom, timestamp=2026-04-28T19:07:17.085, value=sara
row2             column=info:nom, timestamp=2026-04-28T19:08:07.744, value=adam
2 row(s)
Took 0.0351 seconds
hbase:008:0>
```

5. Mise à jour

```
hbase shell
hbase:008:0> put 'etudiants' , 'row1' , 'notes:maths' , '20'
Took 0.0195 seconds
hbase:009:0>
```

6. Suppression

```
hbase shell
hbase:010:0> delete 'etudiants' , 'row1' , 'notes:info'
Took 0.0213 seconds
hbase:011:0> deletall 'etudiants','row1'
Traceback (most recent call last):
NoMethodError (undefined method `deletall' for #<HBaseReceiver:0x44eb2452>)
Did you mean? deleteall
hbase:012:0> deleteall 'etudiants','row1'
Took 0.0087 seconds
hbase:013:0>
```

7. Comptage

```
hbase shell
hbase:010:0> delete 'etudiants' , 'row1' , 'notes:info'
Took 0.0213 seconds
hbase:011:0> deletall 'etudiants','row1'
Traceback (most recent call last):
NoMethodError (undefined method `deletall' for #<HBaseReceiver:0x44eb2452>)
Did you mean? deleteall
hbase:012:0> deleteall 'etudiants','row1'
Took 0.0087 seconds
hbase:013:0> count 'etudiants'
1 row(s)
Took 0.1204 seconds
=> 1
hbase:014:0> 
```

8. Suppression de la table

```
hbase shell
hbase:015:0> disable 'etudiants'
Took 0.7532 seconds
hbase:016:0> drop 'etudiants'
Took 0.4113 seconds
hbase:017:0> 
```